

- The authority key identifier extension SHALL be present.
- The extensions SHALL appear in the same order in the Matter certificate and in the corresponding X.509 certificates.

Note that Matter doesn't specify how firmware images are signed and implementation of firmware image signing is manufacturer-specific. However, since firmware image signing is a common feature, the format for Matter TLV certificates has affordances for encoding firmware signing certificates.

6.5.13. Signature

The context-specific tag **signature** [11] SHALL be used to identify the signature field in the Matter certificate structure.

An **ec-signature** is the encoding of the signature as defined in [Section 3.5.3, “Signature and verification”](#).

```
ec-signature => OCTET STRING [ length (CRYPTO_GROUP_SIZE_BYTES * 2) ]
```

6.5.14. Invalid Matter certificates

The Matter certificate is considered invalid if it violates Matter certificate encoding rules defined in this section. The processing of invalid Matter certificate SHOULD fail and an error SHOULD be reported to the application. Here is a non-exhaustive list of errors that may invalidate the certificate:

- Matter certificate structure includes elements that are not defined in this section.
- Matter certificate elements are encoded in a wrong order.
- Matter certificate element has wrong type.
- Length of an element is outside of its valid range, for example:
 - **serial-num** field is longer than 20 octets.
 - **not-before** or **not-after** field is longer than 32-bits.
 - **subject-key-id** or **authority-key-id** field is different from 20 octets.
- Matter OID values encoded in Matter certificate are not defined in this section, for example:
 - **sig-algo** field encodes value, which is not defined in [Table 84, “Signature Algorithm Object Identifiers”](#).
 - **pub-key-algo** field encodes value, which is not defined in [Table 88, “Public Key Algorithm Object Identifiers”](#).
 - **ec-curve-id** field encodes value, which is not defined in [Table 89, “Elliptic Curve Object Identifiers”](#).
 - **key-purpose-id** field of the Extended Key Usage Extension encodes value, which is not defined in [Table 91, “Key Purpose Object Identifiers”](#).
- Invalid Matter Distinguished Names encoding for Issuer and Subject DNs. Refer to [Section](#)